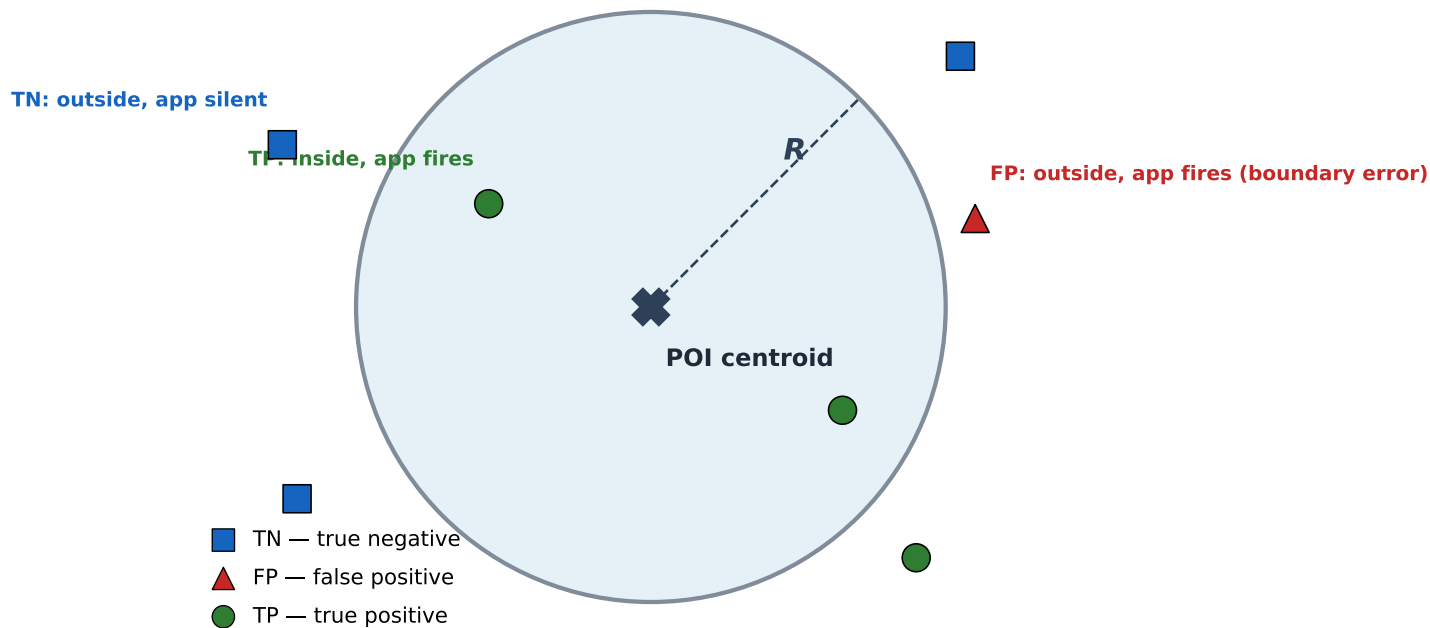
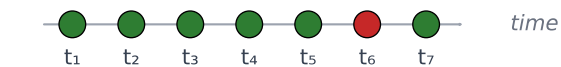


(a) Geofence geometry and trigger labels



(b) Trigger collapse to independent observations

Raw trigger stream during one POI $\times$ radius $\times$ session crossing



7 raw triggers (6 TP, 1 FP) — all within ~ 10 s

collapse rule:
dominant label per
(POI $\times$ radius $\times$ session)

One independent observation

label = TP (dominant)

$n_{\text{raw_triggers}} = 7$

After collapsing: 139 raw triggers $\rightarrow$ 78 independent observations

Wilson 95% CIs and McNemar tests reported on collapsed data.